

## 8086 and 8088 comparison

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| <b>8086</b>                                                                         | <b>8088</b>                                                                      |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Similar EU and Instruction set ; dissimilar BIU</b>                              |                                                                                  |
| <b>16-bit Data bus lines obtained by demultiplexing <math>AD_0 - AD_{15}</math></b> | <b>8-bit Data bus lines obtained by demultiplexing <math>AD_0 - AD_7</math></b>  |
| <b>20-bit address bus</b>                                                           | <b>8-bit address bus</b>                                                         |
| <b>Two banks of memory each of 512 kb</b>                                           | <b>Single memory bank</b>                                                        |
| <b>6-bit instruction queue</b>                                                      | <b>4-bit instruction queue</b>                                                   |
| <b>Clock speeds: 5 / 8 / 10 MHz</b>                                                 | <b>5 / 8 MHz</b>                                                                 |
| <b>In MIN mode, pin 28 is assigned the signal <math>M / \overline{IO}</math></b>    | <b>In MIN mode, pin 28 is assigned the signal <math>IO / \overline{M}</math></b> |
| <b>To access higher byte, <math>\overline{BHE}</math> signal is used</b>            | <b>No such signal required, since the data width is only 1-byte</b>              |

# 8087 Coprocessor

## Multiprocessor system

- A microprocessor system consisting of two or more processors.
- Distributed processing: Entire task is divided in to subtasks

## Advantages

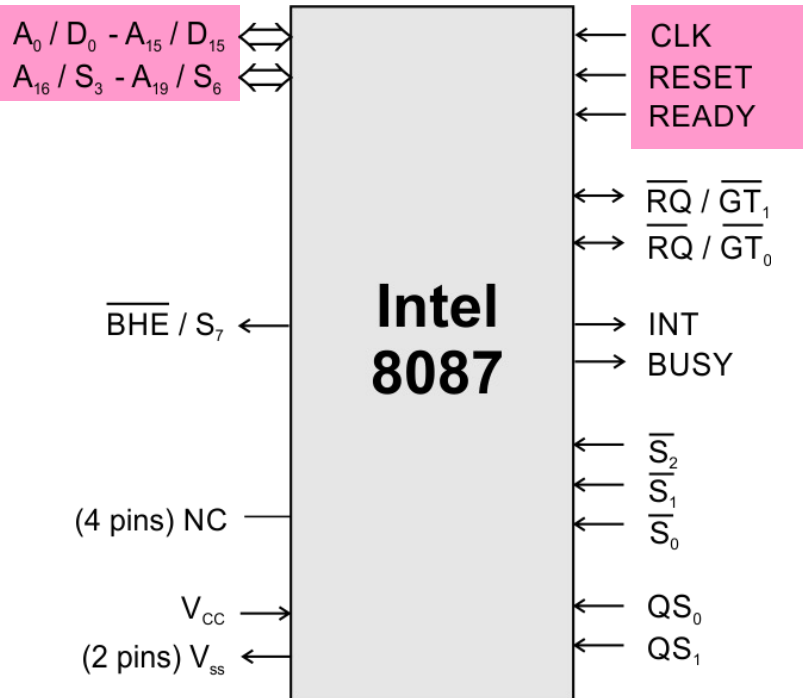
- Better system throughput by having more than one processor
- Each processor have a local bus to access local memory or I/O devices so that a greater degree of parallel processing can be achieved
- System structure is more flexible.  
One can easily add or remove modules to change the system configuration without affecting the other modules in the system

## 8087 coprocessor

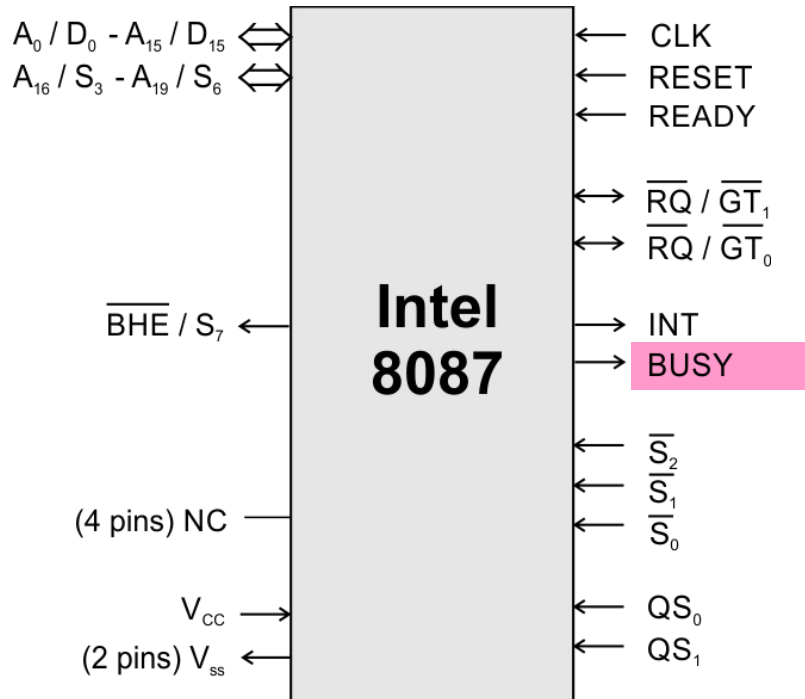
- Specially designed to take care of mathematical calculations involving integer and floating point data
- “Math coprocessor” or “Numeric Data Processor (NDP)”
- Works in parallel with a 8086 in the maximum mode

## Features

- 1) It can work with data of integer, decimal, and real types, with sizes ranging from 2 bytes up to 10 bytes.
- 2) Instruction set involves square root, exponential, tangent etc. in addition to addition, subtraction, multiplication and division.
- 3) High performance numeric data processor  $\Rightarrow$  it can multiply two 64-bit real numbers in about  $27\mu\text{s}$  and calculate square root in about  $36\mu\text{s}$
- 4) Follows IEEE floating point standard
- 5) It is multi bus compatible

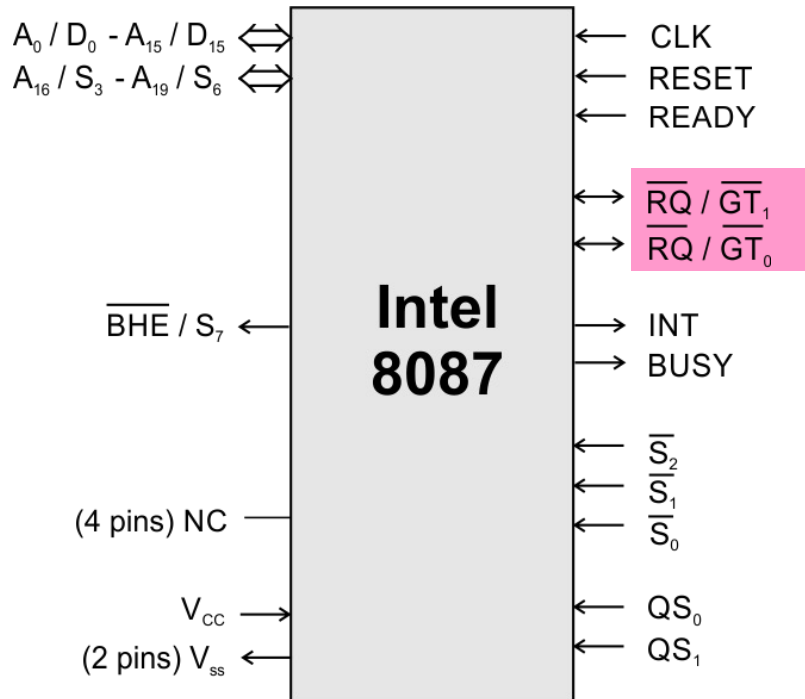


- **16 multiplexed address / data pins and 4 multiplexed address / status pins**
- **Hence it can have 16-bit external data bus and 20-bit external address bus like 8086**
- **Processor clock, ready and reset signals are applied as clock, ready and reset signals for coprocessor**



## BUSY

- **BUSY signal from 8087 is connected to the  $\overline{TEST}$  input of 8086**
- **If the 8086 needs the result of a calculation that the 8087 is still working on, it can use the WAIT instruction to keep checking the TEST pin.**
- **When the BUSY output goes low, it means the 8087 has finished the calculation, and the 8086 can continue**

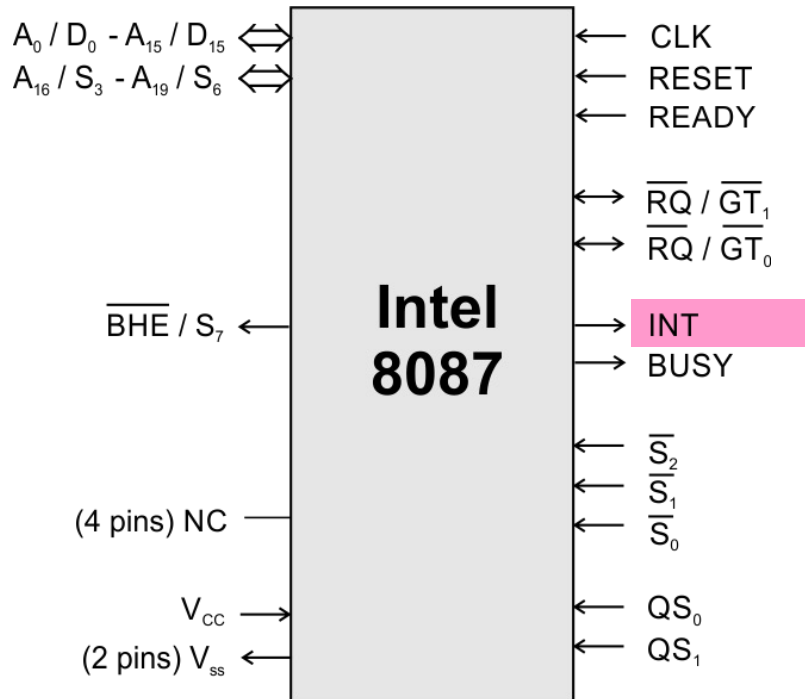


$\overline{RQ} / \overline{GT_0}$

- The request / grant signal from the 8087 is usually connected to the request / grant ( $\overline{RQ} / \overline{GT_0}$  or  $\overline{RQ} / \overline{GT_1}$ ) pin of the 8086

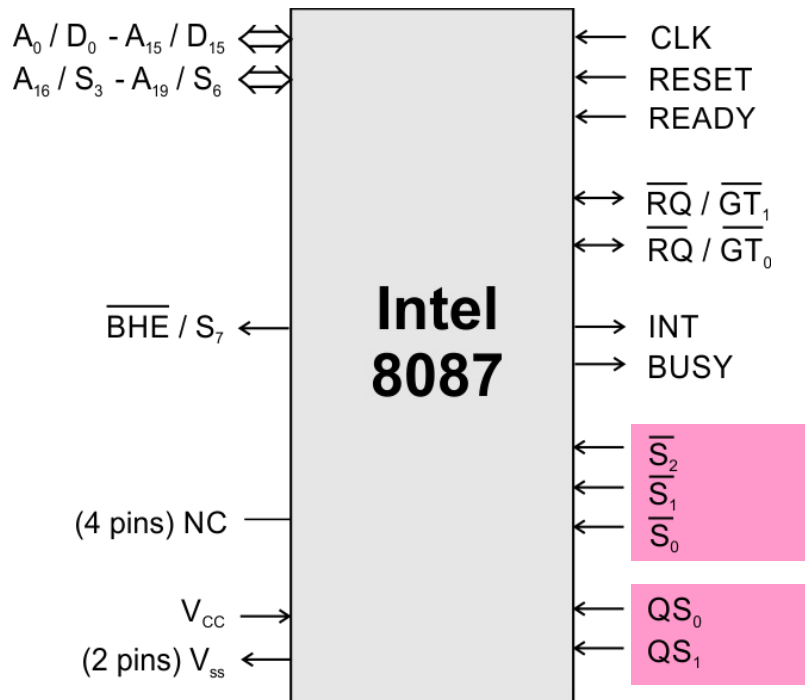
$\overline{RQ} / \overline{GT_1}$

- The request / grant signal from the 8087 is usually connected to the request / grant pin of the independent processor such as 8089



## INT

- The interrupt pin is connected to the interrupt management logic.
- The 8087 can interrupt the 8086 through this interrupt management logic at the time error condition exists

 **$\overline{S}_0 - \overline{S}_2$** 

| $\overline{S}_2$ | $\overline{S}_1$ | $\overline{S}_0$ | Status       |
|------------------|------------------|------------------|--------------|
| 1                | 0                | 0                | Unused       |
| 1                | 0                | 1                | Read memory  |
| 1                | 1                | 0                | Write memory |
| 1                | 1                | 1                | Passive      |

 **$QS_0 - QS_1$** 

| $QS_0$ | $QS_1$ | Status                               |
|--------|--------|--------------------------------------|
| 0      | 0      | No operation                         |
| 0      | 1      | First byte of opcode from queue      |
| 1      | 0      | Queue empty                          |
| 1      | 1      | Subsequent byte of opcode from queue |

## How the 8087 Coprocessor works with the 8086:

1. 8087 instructions are written inside the 8086 program.
  - They look like normal instructions to the 8086.
2. Fetching instructions:
  - Both 8086 and 8087 read the instruction bytes and put them in their own queues.
  - For 8086, unknown 8087 instructions act like **NOP (do nothing)**.
  - 8087 instructions always start with the bit pattern **11011** in their first byte.
3. Instruction checking (done by 8087):
  - 8087 watches the status lines ( $S_2$ – $S_0$ ) and address/data bus ( $AD_0$ – $AD_{15}$ ) of 8086.
  - It also monitors the **queue status** ( $QS_0$ – $QS_1$ ).
    - If  $QS = 00$  → Do nothing.
    - If  $QS = 01$  → 8087 checks the instruction's first 5 bits.
      - If the bits match **11011**, the instruction is for 8087, so it executes it.
      - If there's an error while decoding, 8087 sends an **interrupt request**.
4. During execution:
  - If the instruction needs memory access, 8087 works with 8086 for **read/write operations**.
  - Extra words (operands) are transferred using the **RQ/GT<sub>0</sub>** signal.
  - When 8087 is busy, its **BUSY pin** is set high.
  - 8086 checks the 8087's **TEST pin** and may use the **WAIT instruction** until 8087 finishes.

